

GLOBAL

; up · down · / left/right · **ENTER** select · /DEL back · Home: 2-col grid, letter jumps, // hop columns · { } page · **b** screenshot · **h** help · **Fn+ /Fn+DEL** STOP all radio/rotator control (Help links: **g** glossary+math · **m** user guide · **s** sat history · **t** tech help · **i** learn theory · **f** band plan)

HOME

ENTER opens item; 2-column grid: Satellites, Next Passes, Passes, Track, World Map, Overhead view, Sun/Moon, Space Wx, Activations, AMSAT status, Weather, Grid dist/bearing, Nearby & DX, Location, Update, Settings, Log, Messages, About, Charge/Sleep

SATELLITES

f favorite · **v** favs-only · / search ALL of Celestrak → add as auto-updating fav (10 s/2 h courtesy limits) · **n** new GP sat · **x** del added sat · **e** EQX table · **k** OSCARLOCATOR · **3** 3D globe · **2** sat-to-sat · **o** orbital · **y** sim · **t** transponders · **d** 10-day · **i** illum · **s** AMSAT status · **L** share GP over LoRa · **ENTER** passes · right edge: dot = AMSAT heard, square = telemetry, ring = not heard

SAT-TO-SAT (Sats → 2)

Windows when selected sat + a 2nd fav are BOTH above your horizon (start, dur, peak el each). **n** next sat · **r** recompute · ` back

LIVE GPS POS (Location → v)

DMS lat/lon (max precision) + decimal, alt, grid, speed, course, Rovers/portable. ` back

3D GLOBE (Sats → 3)

Wireframe Earth, auto-follows selected sat. Graticule, coastline, day/night terminator, QTH, favs, selected sat + footprint + ground-track trail. **g** DX 2nd location (grid) + footprint · **G** clear · **arrows** turn · **ENTER** re-follow · ` back

EQX TABLE (Sats → e)

Equator-crossing UTC + longitude (W-positive) for OSCARLOCATOR, next 3 days. **./l** scroll · **d** asc/desc node · **r** recompute · ` back

OSCARLOCATOR (Sats → k)

Live azimuthal-equidistant plot: sub-point, footprint, QTH range ring + full ground track (AOS/LOS). **m** toggle polar (default, auto N/S, flips at equator) / QTH-centered · ` back

DX DOPPLER TABLE (Mutual → ENTER → d)

RX/TX dial freqs for BOTH stations every 30s across a mutual window. Two lines/step: me (green) + DX (cyan). From the Mutual list, **ENTER** opens a polar detail (me+DX arcs, AOS/LOS/el) then **d** the table (or **d** straight from the list), **t** cycle transponder · **m** mode: true rule / fixed DL / fixed UL · **a** anchor dial (me/DX RX/TX) · **./l** in fixed mode step anchored dial to round 1 kHz (else passband 1 kHz) · header shows pb +/- from center · **./l** scroll · ` back

NEXT PASSES (favs)

ENTER track · **m** world map · **t** sky-at-a-glance timeline (bars by elev) · **p** rove planner · **w** workable horizon · **s** target search · **r** refresh · **z** deep-sleep until AOS

WORKABLE HORIZON (Next Passes → w)

10-day union of everything workable across all favorites, **s** states list · **d** DXCC list · **g** re-run incl. grids (off by default) · **w** save to /CardSat/workable/ · ` back

TARGET SEARCH (Next Passes → s)

Pick one state / DXCC / grid; lists every pass over 10 days where it's workable, time-ordered across all favs (sat / date / window / maxEl). **ENTER** polar of that pass · **w** save to /CardSat/search/ · ` back

ROVE PLANNER (Next Passes → p)

Survey all favorites from a planned grid+time. Form: grid / date / time / +/- hrs / GO. Rows by AOS: sat / AOS / maxEl / states / DXCC. **ENTER** detail (arc + **s/d/g** state/DXCC/grid lists) · **w** save txt · **i** saved plans (**ENTER** view, **./l** scroll, **d** del) · **g** form

PASSES (sel)

* = optically visible. **./l** select · **d** detail · **t/ENTER** track · **n** add TX · **r** recompute · **x** mutual-DX · **v** 10-day chart · **V** visible-pass list · **i** illum · **g** grids · **w** US states · **e** DXCC · **p** print

TRACK (sel)

` exits to previous screen & KEEPS radio/rotator tracking (green RAD/ROT/R+R in header); **r/o** to stop. **m** TUNE/CAL · **d** tune mode (FULL/DLU/LFULL/hold; FULLu=OTR on uplink knob) · **t** next TX · **n** jump to beacon · **c** CTCSS · **N** sat note · **k** CW both legs (linear) · **r** radio · **o** rotator · **p** polar · **a** point-here arrow · **z** big readout · **y** tilt on/off (ADV) · **f** Manual · **i** log QSO · **v** voice memo (SD) · **g** grids · **w** states · **e** DXCC now · **ix2** report Heard (AMSAT) · after LOS **q** (60s) deep-sleep to next pass · **ENTER** save cal · **Fn+u** fully release USB (host stays resident after disengage) is when the serial console returns

BIG READOUT (z from Track)

Big RX/TX + az/el + tune mode (follows Track). Radio+rotator keep tracking · **./l** tune · **s/x** step/ctr · **m/d** mode · **t** TX · **n** beacon · **k** CW (linear) · **r** radio · **o** rot · **y** tilt · **i** log · **z/** back

MANUAL (no radio)

Fix one leg, read Doppler freq to tune the other by hand. **u** toggle HOLD/TUNE leg · **./l** passband (linear) · **m** CAL · **t** next TX · **z** big view · **i** log · **p** polar · **g** grids · **w** states · **e** DXCC · **/f** Track

MANUAL BIG (z from Manual)

HOLD/TUNE legs in big digits · **u** swap leg · **./l** tune · **s/x** · **m** CAL · **t** TX · **z/** back

TRACK · TUNE

./l tune **./+** · **s** step 100/1k/5k · **x** recenter · tilt to tune if enabled (ADV)

TRACK · CAL

./l downlink · **./l** uplink · **s** step 10/100/1k · **x** zero

WORKABLE GRIDS / STATES / DXCC

Footprint coverage (per-pass union or live now), count on a cyan line: **g** 4-char grids · **w** US states+DC · **e** DXCC (all 340, hybrid polygons+points). On grids, **f** prefix filter (EM/EM2/EM21, upper-cased), **c** clear. **./l** & { } scroll · ` back

POLAR / PASS DETAIL

Pass detail: **p** polar of this pass. Polar: **i** log QSO · **v** voice memo (SD) · **p/ENTER/** back

SUN / MOON

Graphic sky-dome (Sun/Moon glyphs by az/el) · **g** graphic/list · **./l** pick Sun/Moon · **o** rotor track on/off · **s** sky sources · **t** transits · **e** EME · **x** stop · ` back

SKY SOURCES (Sun/Moon → s)

Planets (cyan dots) + strong radio sources (orange +): Cas A, Cyg A, galactic center, Crab, Virgo A, on a sky dome. Antenna-pointing / RF reference. **./l** select · ` back

EME / MOONBOUNCE (Sun/Moon → e)

Self-echo Doppler per band (50/144/432/1296/10368, topocentric) · range + rate · path degradation vs perigee · galactic sky-noise flag · SUN flag <10° · **m** mutual window · **p** 30-day plan · **o** rotor track Moon · ` back

GRID DIST/BEARING (menu)

Enter Maidenhead grid → great-circle distance + beam heading (short/long path, km/mi). **g** grid · **q** QRZ → grid lookup (seeds calc) · **o** point rotor at bearing · ` back

SPACE WX (menu)

Solar 10.7cm flux · planetary Kp + A index + aurora likelihood (from Kp), labeled & color-coded, with HF/sat operating outlook & data age · **p** HF/6m propagation · **m** MUF to world regions · **r** refresh (WiFi) · **^** back

HF/6m PROPAGATION (Space WX → p)

Turns solar flux + Kp into band guidance: HF conditions (10/15/20m open/marg/shut), geomagnetic effect, auroral-VHF likelihood (6m/2m, beam N), D-layer absorption. Rule-of-thumb · **r** refresh · **^** back

MUF TO REGIONS (Space WX → m)

MINIMUMUF-3.5 path MUF from your QTH to 24 DX World regions: bearing, distance, MUF, best band, color-coded. F-region model, best 800-8000km · **^** scroll · **k** world map (colored dots) · **x** print · **^** back

WEATHER (menu)

Current conditions + multi-day forecast for your site (Open-Meteo). Refreshes on entry (WiFi) & with Update. Units in Settings · **r** refresh · cached offline · **^** back

QRZ LOOKUP (menu)

Callsign lookup via QRZ.com XML (needs QRZ XML subscription + user/pass in Settings → Network). **ENTER** type call → name/address/grid/class. WiFi req'd · **^** back

BAND PLAN (Help → f)

Worldwide amateur band allocation LF→light; HF with ITU R1/R2/R3 splits, VHF/UHF/microwave EME+calling freqs, satellite subbands, IARU designators (H/A/V/U/L/S/C/X/K), sat modes incl QO-100. **^** scroll · **^** back

ACTIVATIONS (menu)

Upcoming sat activations on hams, at (roves, grid/special ops). List: date, call, sat, grid (* = your own entry). **^** move · **ENTER** detail **h** → add your own sked (offline OK), **e** edit, **a** entry · **r** refresh · **^** back. Detail: UTC/mode/freq + **footprint** **note** (checks co-visibility with the activator +/-30 min of listed time). **^** scroll the full comment, a **SKED** reminder (T-60/30/10 beeps+flash), **w** mutual-window screen if a footprint exists. Manual window: small polar plot (me + DX arcs), Date/AOS/LOS/dur + peak el each; **d** DX Doppler pre-set to the transponder + fixed DLUL parsed from the freq/comment (default table if none). Cached to card — last list shows offline; WiFi to refresh

OVERHEAD NOW (menu)

Snapshot of every catalog sat above the horizon right now, sorted by elevation, with az + rise compass dir (high=green, low=yellow) + count up/scanmed · **r** rescan (this instant) · **^** scroll · **^** back

TRANSPONDER DB (Sats → t)

Sat's transponder/beacon entries, two lines each: **D**=downlink+mode line, **U**=uplink+tone/rv line. Ordered two-way > amateur > active; inactive dimmed & **(off)**. **^** select (* = manual) · **x** del manual (2x) · **^** back

ORBITAL ANALYSIS

h 11 pages: Info / Live / Next pass / Ground track / Doppler / Nodal / Sun-Beta / Pass outlook / Orbit position / Phys (velocity + launch date/age) / Explore (what-if apo/per/lncl sandbox, x reseed) · **r** recompute · **z** orbital-zone transits · Doppler **f** sets beacon freq

ORBITAL ZONES (Orbital → z)

When the sat transits distinctive regions, w/ enter/exit times + duration + live in/out + L-shell, z cycle zone: **SAA** / **Eclipse** / **Polar** (>60°) / **Inner** & **Outer Van Allen belts** (L-shell, higher orbits) · **^** scroll · **x** print · **^** back

SIMULATION

h step time · **^** step size · **m** world-map view (sub-point + footprint) · **x** reset to now · **^** back

LOCATION

e/o/a lat/lon/alt · **g** grid · **p** GPS on/off · **s** GPS source · **c** set clock · **v** live position (DMS) · **ENTER** GPS sky plot

GPS SKY PLOT

Live GNSS by az/el, colored by signal (green=strong, gray=weak) · **^** back

WORLD MAP

All footprints + night-side shading · **f** highlight favorite · **c** recenter on QTH/0° · yellow=sunlit cyan=eclipse · **^** back

SCHEDULES

10-day · **^** scroll +/1 day (fills full days), **r** recompute · illum: **h** scroll +/60 days · Mutual: **^** scroll, **ENTER** polar detail, **d** Doppler

LOG

Menu (scrolls): **ENTER** new QSO / browse / export ADIF / LoTW upload / Cloudlog upload / voice memos / notes / awards / fill grids (QRZ) · **Awards**: QSO/grid/date/DXCC totals (states+DXCC deleted from grid), **g/s/d** scrollable worked lists, **ENTER** per-sat · List: **^** scroll, **ENTER** edit · Entry: **^** list, **ENTER** edit, **s** save, **x**2 delete · editing a QSO re-arms its upload; extra **LoTW/Cloudlog** rows (ENTER toggles) override that

SIGN & UPLOAD TO LoTW (Log → Sign & upload)

Uploads sat QSOs direct to ARRL LoTW over WiFi. Needs SD + your LoTW key (lotw_key.pem/lotw_cert.pem in /CardSat/ → make them from your TQSL p12 with tools/lotw_cert_convert.html in a browser, in Settings pick DXCC → primary (state/province...) →

secondary (county/city) + zones + IOTA, all gated pickers w/ type-to-filter) — see manual §8. **u** upload · **a** re-send all · **^** back

UPLOAD TO CLOUDLOG (Log → Upload to Cloudlog)

Uploads sat QSOs to a self-hosted Cloudlog/Upload over WiFi (an alternative to LoTW — Cloudlog can forward to LoTW). Set **URL** + read-write **API** key + **station ID** in Settings. **u** upload · **a** re-send all · **^** back

VOICE MEMOS (Log → Voice Memos)

Browse newest-first (date/time/sat/len). **ENTER** play · **n** new · **d** del · **r** refresh · record via **v** on Track. SD req'd

NOTES (Log → Notes)

Free-form text notes (txt in /CardSat/notes/, newest-first w/ date/time). Browser: **ENTER** open · **n** new · **d**ENTER del. Editor (commands use **F**no **s** /**^** type): type freely, **ENTER**=newline, **DEL**=backspace **F**nt/**F**nv cursor L/R · **F**nt/**F**nv, up/down · **F**nv+s save · **e**xit (auto-saves)

LORA MESSAGES (Home → Messages)

CardSat-to-CardSat broadcast chat (Cap LoRa). Same freq/SF/BW = same group. Set **region** (US 33cm / EU 70cm / JP 430) in Settings for a legal default freq, **n** write/send · **^** scroll+select (newest on bottom) · **o** station roster · **r** retry · **m** LoRa RX/hex monitor · **^** back. **Actionable mssgs** (plain text, interop): a msg with @**lat,lon** + **SAT** / **ISAT** date time → **ENTER** opens bearing/comment (dist/bur/grid) / sat detail / pre-filled sked; **hSAT** carries NORAD (**#name/norad**) so it resolves across differing names. Send for the current sat & your location: (also a presence ping) · **s** satellite · **k** sked (date→time). **Roster** (**o**): who is heard, with grid + dist/bur + signal; **ENTER**=compass, **p**=ping. Opt-in **Auto position reply** setting answers others' positions (look-guarded). **Rx always** on: badge + banner (opt-in beep, **Msg notify** in Settings). **Share GP** elements: a peer's **SAT** → **L** pops an **Imported satellite?** prompt (sender/name/NORAD) — **y** add/update, **n** decline; checksum-verified, experimental. Needs RadioLib build.

LORA RX / HEX MONITOR (Messages → m)

Receive/inspect any LoRa signal (not just sat's). Config: set freq (type in **MHz**), SF, BW, CR, sync, preamble, CRC — saved across reboots. **ENTER** starts RX. Monitor: live hexdump+ASCII, RSS/SNRR, **p** pause (read a frame on a busy channel) · **^** scroll · **s/b/c/t** tune SF/BW/CR/step · **h** freq · **^** esc. Rx-only. Untested HW

UPDATE

kENTER GP+clock+AMSAT+space-wx+weather · **f** fast (GP+AMSAT+favs' TX) · **a** cache all TX (auto-reboots) · **w** WiFi only

SETTINGS

h change · **ENTER** edit/toggle · **s** scan WiFi · opt. 2nd WiFi (field fallback) · **r**eset = ERASE

DUAL RIG (2 radios)

CAT type **Dual** = a downlink + an uplink radio driven natively (27 radios). **Settings → Dual-Rig** setup: per leg pick **Cat Rig**; Bus (Grove / USB / LAN), Cl-V, badge (**Grove honors Cl-V wiring**, incl. 1-pin); a leg set to **None** = that half simply not CAT driven; **LAN** legs take the radio's IP · port (**CT-705** over its own WiFi; **u**p link); a nominate USB adapter (both legs may be USB via a hub) · **s** save+update. One Grove leg only; PTT stays manual; FM tones set on the radio. The CardSatDualRig companion (rigctl net/Grove) is still supported.

GP SOURCE

AMSAT / **CelesTrak** JSON-PP category / **Custom URL** · **^** move · **ENTER** select

ROTATOR (manual)

h az · **^** el · **s** step · **x** stop · **^** back

NETWORK SERVERS

rigctld PC drives rig (VFOA=DLB=UL) · **rotctld** PC drives the configured rotator · **rigctl** drives remote rig · **Web** opt-in mobile page (no auth, trusted LAN); live sky plot. Doppler readout, tap-to-copy freqs, visible-pass list + AOS alerts, radiolrotator control, **Files** = download-only /CardSat browser (no upload)

EDIT

type · **DEL** backspace · **ENTER** ok · **^** cancel

NEARBY & DX (Home)

Live-feeds hub. **APRS** heard: live APRS-IS listen (needs calling; **g** center grid, **ENTER** bearing detail, **f** restart). **DX cluster**: **f** fetch, **n** next band with spots, comments shown. **ADS-B radar**: polar range plot, **f** fetch, **t** scatter target grid; source URL in Settings > Network. **QRZ** lookup, **p** prints each feed.

ABOUT

Build/version, IP, free heap and diagnostics (read-only). **r** Station readiness checklist · **t** **Tools** (63): scientific/graphing/programmer calculators, **Tiny BASIC** (0.9.59: SETSEL/PRINT/gfx; no INPUT), **location converter** (Rig/DMS/DDM/plus/UTM/MGRS), DXCC/CTQ/ITU lookups, RF/antenna wrench, link budget, phasing/split, attenuator, RF exposure, orbit lifetime, **State vector** → GP, Q codes/predictions/RT, CTCSS, **Telnet** (0.9.67: LAN terminal, 10 saved hosts, opt. printing), **0.9.59 satbuild tools**: conjunction scheduler, orbital neighborhood, transponder planner, link-margin curve, debris-group screen, Doppler badge, cascade NFG/GT, sun-noise G/F, helix, LPV/T match, pointing loss, IMD, microstrip Z0, toroid, delta-v, thermal, Faraday, ampacity, PLT plan · **p** **Print**: 30 reports to network printer / serial / 80-cd /Reports file (any mix); contextual **p** on report screens + all form tools, **P** all-passes & polar map · **L** license & credits · **z** **Games menu**: seven mini-games (g, **pick**, **ENTER** launch).